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data communication (Pokhara University)

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POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Data Communication

Semester: Fall

Year : 2013
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

a) Explain the working of data communication system with block diagram. What are the merits of digital data transmission over analog data transmission? 7

b) Differentiate between 8

i. Synchronous and asynchronous communication

ii. Parallel And serial communication.

a) Why is data collision protocol necessary? How CSMA/CD detects and handles collision during data transmission? 7

b) Determine if the system described by y(t)=cos[x(t+5)] is casual, linear, time variant and stable. 8

a) What is error in data transmission? Explain the operation of Checksum for error detection purpose with an example. 7

b) What is Unit Step function and impulse function? Obtain the fourier transform of Unit Impulse function with suitable conclusion on result. 8

a) Find the fourier transform of X(t) = 1 for 0<t<T 5
0 elsewhere . Also draw the waveform of X(t) .

b) Explain various types of propagation of electromagnetic wave. 5

c) Explain the characteristics of guided transmission methods. 5

a) What is biphase encoding? How does biphase overcome the limitation of NRZ codes? Explain with examples. 7

b) Describe flow control and error control. How does Go-back-N ARQ work? 8

a) List various types of data compression techniques with their brief description. 5

b) Differentiate encoding and modulation. 5

c) Define multiplexing. What are its applications? 5

7. Write short notes on: (Any two) 2×

a) Power versus Energy signal.

b) Message Switching.

c) Standard of modem.

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POKHARA UNIVERSITY

Level: Bachelor
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Semester: Spring

Year : 2013
Full Marks: 100
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Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

a) What are the data communication terminologies? Write about evolution of data communication system. 8

b) Illustrate a situation when baud rate can be lower than bit rate. Also discuss the function of RS-232C interface in brief. 7

a) Differentiate energy and power signal. Is the system y(n)=n.x(n) is 7
   i. Linear
   ii. time variant

b) Sketch the given signal and obtain its fourier series. 8

$X(t) = 0 , -T/2 < t < -T/4$

$A , -T/4 < t < T/4$

$0 , T/4 < t < T/2$

a) Briefly explain TCP/IP layers and compare it with OSI reference model? 7

b) Explain the importance of error control and flow control in communication system. Briefly explain sliding window flow control technique. 8

a) What are guided and unguided transmission media? Discuss the advantages of fiber optic over other guided media. 8

b) State various data compression techniques used in data communication. Explain one of them in detail. 7

a) What is the necessity of modulation? Draw and explain the block diagram of PCM system. 7

b) Explain the term multiplexing and switching. What are the components of Private Branch Exchange. 8

a) Which frequency ranges are of interest in unguided transmission media? Explain the different modes of optical fiber transmission. Point out the advantages of optical fiber over other guided media. 8

b) Differentiate between encoding and modulation. Encode the bit stream 1100001011 using 7
   i. Polar RZ
   ii. Polar NRZ
   iii. AMI
   iv. Manchester
   v. HDB3

7. Write short notes on: (Any two) 2×
   a) DPSK
   b) Unipolar and Bipolar encoding
   c) Unit step function and impulse function.

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POKHARA UNIVERSITY

Level: Bachelor
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Semester: Fall

Year : 2014
Full Marks: 100
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Time : 3hrs.

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Attempt all the questions.

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a) Differentiate between analog and digital data transmission, which one will you prefer for long distance transmission and why? 7

b) Differentiate between 8

i. Synchronous and asynchronous communication

ii. Serial and parallel transmission.

c) Define system. For the system described by characteristics equation: $y(t) = Ax(t) + B$, determine whether the system is, 7

i. Linear or non-linear

ii. Casual or non-casual

iii. Stable or unstable

iv. Shift variance or shift invariance

v. Static or dynamic.

d) Obtain the fourier series of following signal. 8

$$X(t) = -A, for -\frac{T}{2} < t < -T/4$$

$$+A, for -\frac{T}{4} < t < +T/4$$

$$-A, for +\frac{T}{4} < t < +T/2$$

2. a) What is ALOHA and what are its drawbacks? How CSMA/CD overcomes it? Explain. 7

b) Compare the characteristics of twisted-pair cable and co-axial cable. Explain different propagation modes in unguided communication. 8

3. a) What is redundancy? Explain the operation of CRC for error detection purpose with an example. 7

b) Differentiate between FDM and TDM. Explain the advantages and disadvantages of packet switching. 8

5. a) Why is stop-and-wait ARQ inefficient? Explain the major difference between Go-back-N ARQ and selective-reject ARQ. 7

b) Differentiate between pulse code modulation and delta modulation. 4

c) Explain the need for multiplexing and switching in data communication. 4

6. a) Differentiate encoding and modulation. The sequence of the bit is 010111100. What will be the output if following encoding schemes are used? 8

i. AMI

ii. NRZ

iii. NRZ-L

iv. RZ

v. Differential Manchester.

b) What is unit step function and unit impulse function? Obtain the Fourier transform of unit impulse function. 7

7. Write short notes on: **(Any Two)** 2×

a) RS-232C Interface Standard

b) Encoding Vs. modulation

c) QPSK modem.

POKHARA UNIVERSITY

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Semester: Spring

Year : 2014
Full Marks: 100
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Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

a) What is data communication? Explain the data communication block diagram with an example. 8

b) What are the basic system properties? Explain. Determine whether x (t) = $5t^2$ is linear or not. 7

a) Check whether x(t)=Acos ωt is power or energy type signal. 7

b) Calculate the Fourier transform of: 8

i. Unit step function.

ii. Signum function.

a) Describe each layer of OSI reference model. Compare it with TCP/IP. 8

b) Define antenna. Explain different types of antenna. 7

a) Explain different types of signal impairments in communication system. 8

b) What is ARQ? How does stop and wait ARQ and sliding window ARQ deals with error. 7

a) What is packet switching? How the input circuit can be multplixed in analog hierarchy. 7

b) What is PBX? List the advantage of PBX in communication. 8

a) Define Modulation. Explain the use of modulation. 5

b) What is DC wonder? Using HDB3 technique generatesthe signal response for this code 11000011100001. 5

c) Define modem and explain the High speed modem(56 modem). 5

Write short notes on: (Any two) 2×5

a) Line configuration.

b) Energy Vs. Power signal.

c) Cellular Telephoy.

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POKHARA UNIVERSITY

Level: Bachelor Semester:Fall Year : 2015
Programme: BE Full Marks: 100
Course: Data Communication Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

a) Describe the block diagram of data communication system. How is digital transmission superior to analog transmission system? 8
b) Discuss different types of serial data transmission .Also illustrate a situation when baud rate can be lower than bit rate. 7
a) Define energy and power signals. Explain basic system properties with example of each. 7
b) Find the Fourier transform of unit step function. Why mathematical tool like Fourier series and Fourier transform are used in engineering? 8
a) Which multiple access technology is used in Ethernet? How does this detect and handle collision during data transmission? Explain. 7
b) Explain basic network characteristics of x.25, frame relay and ATM? 8
a) What are guided and unguided transmission media? Discuss the advantages of fiber optic over other guided media. 8
b) How is burst error difficult to handle than single bit error? Explain hamming code of error correction. 7
a) Why is flow control necessary? Explain sliding window flow control mechanism. 7
b) What is significance of switching? Explain different multiplexing techniques briefly with application of each. 8
a) How does BPSK differ from DPSK? How does Delta modulation overcome the demerits of PCM? Explain. 8
b) Differentiate between encoding and modulation. Encode the bit stream 1110000101 using: 7
   i. Polar RZ.
   ii. Polar NRZ.
   iii. AMI.
   iv. Manchester.
   v. HDB3.
7. Write short notes on: (Any two) 2×
   a) Cellular Telephony.
   b) QPSK modem.
   c) Analog hierarchy.

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POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Data Communication

Semester: Spring

Year : 2015
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Write down the advantages of digital transmission over analog transmission. 8

   A TV channel has a signal with bandwidth of 12 MHZ & it is transmitted with the average power of 6W. This signal has to travel through a channel which introduces an average noise of 0.1mW. If this TV signal is digitized & sent, find the maximum data rate of the channel.

   b) Explain the differences between synchronous and asynchronous serial communication. Define energy and power signals. 7

2. a) Define system. Check this system Y(n)= nX(n) it linear and causal or not. 7

   b) Explain unit impulse function.Calculate the fourier transform ofx (t) =A+2. 8

3. a) Describe each layer of OSI reference model. Compare it with TCP/IP. 8

   b) Explain the working principle of optical fibre. List the advantages over other transmission medias. 7

4. a) What is error control?Explain how hamming code of error correction works with example. 7

   b) Explain how HDB3 and B8ZS methods to handle long runs of zeros which AMI fails to handle. Explain Phase modulation. 8

5. a) Differentiate between circuit packet and message switching. 8

   b) How do we represent ASK and FSK modulation?Represent 1010101 using Manchester and AMI code. 7

6. a) Define modem band rate and explain the 56 modem. 7

   b) With detailed figures explain QPSK technique. 8

7. Write short notes on: (Any two) 2×

   a) VSAT

   b) Private Branch Exchange

   c) Data compression.

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POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Data Communication

Semester: Fall

Year : 2016
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable
The figures in the margin indicate full marks
Attempt all the questions.

a) Explain the applications of data communication. Explain different serial communication modes. 8
b) Define Baud Rate. Explain RS 232C Standards. 7
c) Explain the benefits we get when we replace analog transmission with digital transmission. 7
Check causality and time-invariance of a signal:
$y(t)=x(t)+5u(t-5)$
b) Define LTI systems. Explain significance of frequency domain analysis of signals. 8
3. a) How will you determine whether a signal is power signal or energy signal? Explain. Find fourier transform of unit impulse signal. 7
b) Define network Topology. Explain CSMA/CD with necessary diagrams. 8
4. a) Explain CRC method for detection and correction of error. Explain how data link handles lost and duplicate frames that carry data acknowledgements. 7
b) Explain the advantages and disadvantages of Stop-and-wait ARQ. Explain wave-division multiplexing. 8
5. a) Define multiplexing. Differentiate between Circuit Switching and Packet Switching. 7
b) What do you mean by modulation and demodulation? Explain FM. 8
6. a) Explain different methods to convert analog signal into digital data. 7
b) Explain Delta Modulation. Enumerate QPSK Modem in detail 8
7. Write short notes on: (Any two) 2×5
a) Data Compression
b) PBX
c) VSAT

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Level: Bachelor | Semester: Spring | Year : 2016
Programme: BE | Full Marks: 100
Course: Data Communication | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

a) Explain the working of data communication system with block diagram. What are the merits of digital data transmission over analog data transmission? 7

b) Differentiate between 8
   i. Synchronous and asynchronous communication
   ii. Serial and parallel transmission

a) Define channel capacity with necessary theorem. For a RF signal with band width 6MHz and it is transmitted with average power 2W where 1.2mW noise signal is present on it. If this signal is digitized and sent find the maximum data rate of the channel. 7

b) Find Fourier series representation of a signal: 8
$x(t)=|A| \quad -T/2<t<T/2$

a) Explain unit step function and impulse function. Obtain the Fourier transform unit impulse signal. 7

b) Compare the characteristics of twisted-pair cable and co-axial cable. Explain different propagation modes in unguided communication. 8

a) What is ALOHA and what are its drawbacks? How CSMA/CD overcomes it. Explain. 7

b) Define error with its types and explain the error detection techniques. 8

a) Explain cellular telephony. Explain ground, sky and line-of-sight propagation. 8

b) Explain TDM. Compare Virtual-circuit network with Datagram network. 7

a) Differentiate encoding and modulation. The sequence of the bit is 01011100. What will be the output if following encoding schemes 8

   i. AMI
   ii. NRZ
   iii. NRZ-L
   iv. RZ
   v. Differential Manchester

b) What is a modem? Describe multilevel modulation. 7

7. Write short notes on: (Any two) 2×5
   a) Encoding Vs Modulation
   b) RS-232C Interface Standard
   c) VSAT

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Level: Bachelor
Programme: BE
Course: Data Communication

Semester: Spring

Year : 2017
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

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Attempt all the questions.

a) Explain digital communication system with its block diagram. 8

b) Define Baud rate. An analog signal has a bit rate of 8000 bps and a baud rate of 1000 baud. How many data elements are carried by each signal element? How many signal elements do we need? 7

c) Differentiate between energy and power signal with example. Justify whether $x(t)= e^{-at}u(t)$ (a>0) is energy signal or power signal 8

d) Define LTI system. Find output of LTI system with input signal $x(t)=u(t)$ and impulse response $h(t)=u(t)$. 7

e) Find Fourier series coefficients of signal $x(t)=1+\sin w_o t+\cos w_o t+\cos(2w_o t+\pi/4)$. 7

f) Why we need Fourier transform even we have Fourier series? Find Fourier transform of periodic signal $x(t)=\sin(0.5t)+\cos(0.5t)$. 8

g) Describe X.25, Frame Relay and ATM briefly. 7

h) Compare twisted pair cable, co-axial cable and optical fiber based on frequency range, attenuation, delay and repeater spacing. 8

i) What is redundancy? What are different types of errors? Explain their detection and correction techniques. 8

j) What is ARQ? What are the two types of sliding window ARQ error control? Explain. 7

k) What is multiplexing? Explain FDM hierarchy in telephone system. 7

l) An audio signal of $10\sin 1000\pi t$ is used for AM with a carrier of $50\sin 200000\pi t$. Calculate: 8

i. Modulation Index
ii. Required bandwidth
iii. Total power

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Level: Bachelor Semester: Fall Year : 2017
Programme: BE Full Marks: 100
Course: Data Communication Pass Marks: 45
Time : 3hrs.

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The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Describe the block diagram of data communication system. Write unique features of digital transmission system.
   b) Differentiate parallel and serial transmission. Explain different modes of serial transmission.
2. a) Explain Null-modem configuration in RS-232C. Explain the relationship between baud rate and bit rate.
   b) Explain maximum channel capacity of noiseless and noisy channel.
3. a) Define energy and power signals. Explain why a sinusoidal wave is periodic and deterministic signal.
   b) Explain unit step function and impulse function. Obtain the Fourier transform of unit impulse signal.
4. a) Compare the characteristics of twisted-pair cable and co-axial cable. Explain the working principle of satellite communication.
   b) Explain Frame Relay. Write the advantages of layered network model.
5. a) How is burst error difficult to handle than single bit error? Explain why CRC is better than parity method of error detection.
   b) Why is flow control required in data communication? Explain how lost and duplicate packets are handled by selective repeat request ARQ.
6. a) Differentiate between FDM and TDM. Compare Virtual-circuit network with Datagram network.
   b) Explain phase modulation and frequency modulation.
7. Write short notes on: (Any two)
   a) Response of a system
   b) Lossless and lossy compression
   c) QPSK modem

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POKHARA UNIVERSITY

Level: Bachelor Semester: Fall Year : 2018
Programme: BE Full Marks: 100
Course: Data Communication Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) On a transmission channel, 600 characters message using 7 bit code is used. For sync data stream two sync characters and single error detection is added. In asynchronous data transmission, there is one start bit, one stop bit and error detection character is added. Calculate the efficiency of transmission in both transmission modes. 7
   b) Differentiate between energy and power signal with example. Justify whether $x(t)= e^{-a|t|}$ $(a>0)$ is energy signal or power signal. 8
2. a) What is unit step function and impulse function to obtain the Fourier transform of unit step function? 7
   b) Why we need Fourier transform even we have Fourier series? Find Fourier transform of periodic signal $x(t)=\sin(0.5t)+\cos(0.5t)$. 8
3. a) Explain TCP/IP layers and compare it with OSI reference model. 7
   b) Compare twisted pair cable, co-axial cable and optical fiber based on frequency range, attenuation, delay and repeater spacing. 8
4. a) How error can be defined? Explain different error detection techniques. 7
   b) Explain the importance of flow control. Briefly explain sliding window flow control mechanism. 8
5. a) What is multiplexing? Explain FDM hierarchy in telephone system. 7
   b) Differentiate between encoding and modulation. With diagram, explain QPSK technique. 8
6. a) What are the different techniques of multiplexing? 7
   b) An audio signal of $10\sin 1000\pi t$ is used for AM with a carrier of $50\sin 200000\pi t$. Calculate: 8

   i) Modulation Index
   ii) Required bandwidth
   iii) Total power
   iv) Efficiency of AM
7. Write short notes on: (Any two) 2×5
   a) VSAT
   b) Digital Data Transmission
   c) Private Branch Exchange

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POKHARA UNIVERSITY

Level: Bachelor Semester: Fall Year : 2019
Programme: BE Full Marks: 100
Course: Data Communication Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What are the advantages of Digital data communication over Analog data communication? Draw and explain the generic block diagram of data communication system. 7
   b) Define bit rate and Baud rate. Explain the RS-232C standard with handshake signals. 8
2. a) How do you classify the signal? Differentiate energy and power signal. 7
   b) Explain OSI reference model as a layered architecture with function of each layer. 8
3. a) Explain with example how Fourier Transform helps on the analysis of signal. 8
   b) Define unit impulse and unit step function. Obtain the Fourier transform of a single sided exponential function of e^{-at}. 7
4. a) What do you mean by propagation? Why do we need it? Explain with different propagation mechanism. 7
   b) Explain how can error be controlled using stop and wait methods and sliding window method. Also mention the advantages and disadvantages for both the methods. 8
5. a) Define impairments. What are the basic signal impairments in communication system? Explain. 8
   b) Differentiate encoding and modulation. The sequence of the bit is 01010000011. What will be the output if following encoding schemes are used? 7
      i. AMI
      ii. NRZ-L
      iii. Binary 3 Zero Substitution (B3ZS)

      iv. Differential Manchester.
6. a) Write notes on Analog services and its Hierarchy? Compare and contrast different types of switching mechanism. 8
   b) What is multilevel modulation? Explain QPSK modulation technique. 7
7. Write short notes on: **(Any two)** 2×5
   a) Continuous and Discrete Time signal
   b) Unguided Transmission Media
   c) Private Branch Exchange

POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2019
Programme: BE | Full Marks: 100
Course: Data Communication | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

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Attempt all the questions.

1. a) Explain data frame format of synchronous and asynchronous communication and differentiate them. 7

   b) Check whether the given system is linear, casual and time invariant or not. 8

   i. y(t)=log x(t)

   ii. y(t)=t*x(t)

2. a) What is the importance of using mathematical tool like Fourier Series and Fourier Transform in engineering? Find the Fourier transform of signum function. 7

   b) Explain TCP/IP model with the protocols associated with each layers. Also explain the concept of addressing in each layers of this model. 8

3. a) Explain the electromagnetic spectrum and its application in wireless transmission media with their modes of propagation. 7

   b) What is the purpose of hamming code?Consider the following message M = 1010001101. The cyclic redundancy check (CRC) for this message using the divisor polynomial $x^5 + x^4 + x^2 + 1$ calculate error free message at receiver. 8

4. a) Define unit step signal and unit impulse signal. Prove that convolution in time domain corresponds to multiplication in frequency domain. 4+4

   b) Define flow control and error control. Explain how stop-and –wait ARQ protocol operate. 7

5. a) Describe the goals of multiplexing. Differentiate between FDM and TDM. Assume that a voice channel occupies a bandwidth of 4 KHz. We need to multiplex 10 voice channels with guard bands of 500 Hz using FDM. Determine the bandwidth of the link. Also show the multiplexing scheme diagrammatically. 7

   b) Encode the bit stream 10110001110 using the following scheme. 8

   i. RZ

   ii. NRZ-I

   iii. NRZ-L

   iv. AMI

   v. Manchester

6. a) An audio signal of $20\sin 2\pi.500t$ is used for AM with a carrier of $60\sin 2\pi.10^6 t$. Calculate 6

   i. Modulation index

   ii. Required bandwidth

   iii. Total power using load resistance of 80ohm

   iv. Efficiency of AM

   b) Why delta modulation is preferred over pulse code modulation? 5

   c) Differentiate between circuit switching and packet switching. 4

7. Write short notes on: (Any two) 2×5

   a) Models of a data communication

   b) Cellular telephony

   c) Energy Vs Power Signal

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Comm - 6th

POKHARA UNIVERSITY

Level: Bachelor Semester: Fall Year : 2020
Programme: BE Full Marks: 100
Course: Data Communication Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Explain the evolution of data communication. Draw and explain in brief a model of a data communication system. 8
   b) What is serial and parallel data transmission? Differentiate between synchronous and asynchronous data transmission. 7
2. a) Comment about the linearity, stability, time-invariance and causality for the following system: $y(t)=t^2 x(t)$ 8
   b) Why OSI layer is called referenced model. Explain session layer and presentation layer in brief. 7
3. a) Justify "Fourier transform is essential in data communication" with relevant examples. Find Fourier Transform of the signal e^{-at}. Plot magnitude and phase response. 4+7
   b) Compare Impulse function and impulse response. 4
4. a) What are different modes of propagation in wireless communication? Explain merits and demerits of optical fiber. 8
   b) Explain how redundancy is introduced in parity method of error detection and correction 7
5. a) Explain the need of data compression in data communication. 8
   How is data link error control different from flow control? Explain
   b) Explain the multiplexing technique applied in digital telephony with their significance, applications and multiplexing hierarchy. 7
6. a) What is Line Coding? Represent the given sequence of bits 111000010 using: 6
   i. AMI
   ii. HDB3
   iii. NRZ

   b) Explain PCM as an A to D conversion technique with diagram. 5
   c) Compare encoding and modulation technique. 4
7. Write short notes on: **(Any two)** 2×5
   a) Circuit, message and packet switching
   b) Energy and power signal
   c) VSAT

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POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Data Communication

Semester: Fall

Year : 2021
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Draw a generic digital model of a data communication system and explain in brief the basic signal processing operations at each block. 7

   b) Explain the concept of bit rate, baud rate. Explain data frame format of synchronous and asynchronous communication. 8

2. a) Find the energy and power of signal $x(t)=e^{-4t}.u(t)$ and justify whether it is energy signal or power signal. 8

   b) What is a system? How would you say that a system is BIBO stable? Comment about the stability for the following system: 7
   $y(n)=2x(n+1) + [x(n-1)]^2$

3. a) Why we need Fourier transform even we have Fourier series? Find Fourier transform of unit step signal and signum signal. 8

   b) OSI layer architecture is a reference model. Support this statement and explain the basic function of each layer. 7

4. a) Compare coaxial cable, twisted pair cable and optical fiber with their performance. 8

   b) Explain different methods for the detection and correction of single bit error and burst error. 7

5. a) What is ARQ? What are the two types of sliding window ARQ error control? Explain. 8

   b) Explain the multiplexing technique applied in digital telephony with their significance, applications and multiplexing heirarchy. 7

6. a) What is line coding? Represent the given sequence of bits 1111000011 using: 7
      i. NRZ-I
      ii. Differential Manchester
      iii. HDB3

   b) How an analog signal is converted into digital? Explain QPSK as an multilevel modulation with necessary diagram. 8

7. Write short notes on: **(Any two)** 2×5
   a) Optical fiber Communication
   b) QAM modulation
   c) Circuit switching Vs packet Switching

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2. a) Find the energy and power of signal $x(t)=e^{-4t}.u(t)$ and justify whether it is energy signal or power signal. 8

   b) What is a system? How would you say that a system is BIBO stable? Comment about the stability for the following system: 7

   $y(n)=2x(n+1) + [x(n-1)]^2$

3. a) Why we need Fourier transform even we have Fourier series? Find Fourier transform of unit step signal and signum signal. 8

   b) OSI layer architecture is a reference model. Support this statement and explain the basic function of each layer. 7

4. a) Compare coaxial cable, twisted pair cable and optical fiber with their performance. 8

   b) Explain different methods for the detection and correction of single bit error and burst error. 7

5. a) What is ARQ? What are the two types of sliding window ARQ error control? Explain. 8

   b) Explain the multiplexing technique applied in digital telephony with their significance, applications and multiplexing heirarchy. 7

6. a) What is line coding? Represent the given sequence of bits 1111000011 using: 7

   i. NRZ-I
   ii. Differential Manchester
   iii. HDB3

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b) How an analog signal is converted into digital? Explain QPSK as an multilevel modulation with necessary diagram. 8

7. Write short notes on: (**Any two**) 2×5

a) Optical fiber Communication

b) QAM modulation

c) Circuit switching Vs packet Switching

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POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2021
Programme: BE | Full Marks: 100
Course: Data Communication | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Describe the block diagram of data communication system. How is digital transmission superior to analog transmission system? 8
   b) Differentiate parallel and serial transmission. Explain different modes of serial transmission. 7
2. a) Explain Null-modem configuration in RS-232C. Explain the relationship between baud rate and bit rate. 8
   b) State the properties of Continues time fourier series. 7
3. a) Define energy and power signal. Determine whether the signal $x(t) = A[u(t+a) - u(t-a)]$ for a>o is a Power signal or energy signal. 8
   b) For a given signal $x(t) =$ coswt. Find the magnitude and phase spectrum. 7
4. a) Differentiate between OSI layer and TCP/IP layer. State the role of CSMA/CD in data communication networking. 7
   b) Compare guided transmission media in terms of bandwidth, channel capacity, SNR, security and cost. 8
5. a) Explain different types of data compression techniques in data communication. Highlight its importance in communication. 8
   b) Compare asynchronous and synchronous data-link protocols. 7
6. a) Differentiate between AM and FM. Why is FM superior over AM in communication? 8
   b) Draw ASK, FSK and PSK of the given binary sequence: 0100110011001010. 7
7. Write short notes on: (**Any two**) 2×5
   a) Satellite Communication
   b) Packet switching and message switching
   c) ATM and Frame Relay

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POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2022
Programme: BE | Full Marks: 100
Course: Data Communication | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Why data communication is important? Draw and explain a generic data communication model. 7

   b) Differentiate between different types of transmission modes. Briefly explain RS-232C. 8

2. a) Define LTI system. Explain with examples, how periodic and non periodic signals are distinguished. 8

   b) Check whether $x(t)=A\cos \omega t$ is power or energy type signal. 7

3. a) Define discrete time Fourier series. Find the Fourier transform of unit impulse function. 8

   b) List the different types of network and explain its topology. 7

4. a) Why wireless propagation is important? Explain types of propagation. 7

   b) Enlist the advantages of optical fibre? Explain working principle of optical fibre. 8

5. a) Why burst error is difficult to handle than single error? Explain why CRC is better than parity method of error detection. 7

   b) Why flow control is needed? Explain sliding window flow control algorithm in detail. 8

6. a) Compare and contrast between circuit switching, packet switching and message switching. 8

   b) What is modulation? Explain QPSK modulation in detail. 7

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7. Write short notes on: (**Any two**) 2×5

a) Baud Rate

b) PCM

c) Modems

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POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2022
Programme: BE | Full Marks: 100
Course: Data Communication | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What are standards? Explain its types. Also explain the standard organizations in data communication. 8

   b) Define bit rate and baud rate? Explain with a scenario to show bit rate is not always equal to baud rate with suitable diagram and calculations. 7

2. a) Define Time Invariant System. Explain the following properties of the systems: a) causality b)linearity c)stability d)memory 8

   b) Explain deterministic and random signal with example. Justify whether unit step signal is energy signal or power signal 4+3

3. a) Explain the existence of Fourier Transform. Find the Fourier Transform of Signum function. 7

   b) Compare and Contrast Frame relay, X.25 and ATM. 8

4. a) Why optical fiber is considered advantageous over other copper media? Explain. Also, explain about different propagation techniques. 8

   b) A bit stream is 10011101 is transmitted using the standard CRC method. The generator polynomial is x^3+1. Derive the actual bit string transmitted. Show that the error is detected if any one bit in the received bit stream is inverted. 7

5. a) Define flow control as a layered protocol model. Explain the mechanism of a flow control for an ARQ model. 8

   b) Compare and Contrast between Virtual Circuit Network and Datagram network. 7

6. a) Differentiate between encoding and modulation? Explain differential PSK with an example. 7

   b) How modulation helps to reduce the size of antenna, illustrate in brief. Explain about, amplitude, frequency and phase modulation. 8

7. Write short notes on: **(Any two)** 2×5

   a) Synchronous and asynchronous Communication

   b) Types of Switching

   c) TCP/IP VS OSI Model

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POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Data Communication (Old)

Semester: Fall

Year : 2023
Full Marks: 100
Pass Marks: 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Describe the block diagram of data communication system. Write unique features of digital transmission system. 4+4
   b) Differentiate parallel and serial transmission. Explain different modes of serial transmission. 3+4
2. a) Differentiate between energy and power signal with example. Justify whether $x(t)= e^{-a|t|}$ $(a>0)$ is energy signal or power signal. 7
   b) Comment about the causality, linearity, stability, time variance of a continuous time system given by: $\mathbf{y(t)= ax(t) +b}$ 8
3. a) What is Unit Step Function and Impulse Function? Obtain the Fourier transform of Unit Impulse. Also comment on the result. 7
   b) What is PDU? Differentiate between LLC and MAC Sublayer of OSI reference model. 8
4. a) Compare the characteristics of twisted-pair cable and co-axial cable. Explain the working principle of satellite communication. 4+3
   b) Why is flow control required in data communication? Explain how lost and duplicate packets are handled by selective repeat request ARQ. 8
5. a) Explain the importance of flow control. Briefly explain sliding window flow control mechanism. 8
   b) Differentiate between encoding and modulation. With diagram, explain QPSK technique. 7
6. a) Differentiate between PSK, ASK and FSK modulation technique. with necessary diagram. 8
   b) What is multiplexing? Explain FDM hierarchy in telephone system. 7
7. Write short notes on: **(Any two)** 2×5
   a) Packet Switching and message switching.
   b) Types of noise.
   c) Bit Rate/Baud Rate.

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POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2023
Programme: BE | Full Marks:100
Course: Data Communication (New) | Pass Marks: 45
Time :3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define Simplex, Half Duplex and Full Duplex mode of data transmission. Briefly explain digital communication system with the help of general block diagram. 8
   b) Differentiate between different types of transmission modes. Briefly Explain RS- 232C. 7
2. a) Define Signal. Briefly explain periodic and non-periodic signals, energy and power signals, deterministic and random signals, continuous and discrete time signals. 7
   b) Define unit step signal with necessary mathematical expression and graphical illustration. Check the following system for linearity, causality, static & dynamic and time invariance. 8

$$y(n) = Ax(n) + B$$

3. a) Compare and contrast between OSI and TCP/IP with necessary figures. 7
   b) Define guided and unguided transmission media. State the advantages of optical fiber over other guided media. 8
4. a) Why burst error is difficult to handle than single error? Explain why CRC is better than parity method of error detection. 7

OR

Explain various transmission impairments of a communication system.

   b) Compare and contrast between lossless and lossy compression. 8
5. a) Explain the importance of flow control. Briefly explain sliding window flow control mechanism. 8
   b) Explain the multiplexing technique applied in digital telephony with their significance, application and multiplexing hierarchy. 7
6. a) Is switching and routing same? Differentiate between message switching, circuit switching and packet switching. 8

   b) Differentiate between PSK, ASK and FSK modulation techniques with necessary diagram. 7

OR

Encode the Bit Stream 011010001 using the following scheme.

   i. NRZ – I
   ii. NRZ _ L
   iii. Manchester
   iv. Differential Manchester

7. Write short notes on: **(Any two)** 2×5
   a) Cellular Telephony
   b) Point to Point protocol
   c) QPSK

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